

Muhammad Umer Sajid

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Electrical Engineering finalist at **NUST** (CGPA **3.79/4.00**) with **2+ years** of concurrent industry experience in production-grade embedded firmware, high-speed PCB design, signal integrity, and mixed-signal systems. Proficient in **STM32**, **ESP32**, **Renesas**, **RF/SDR** platforms, and medical-device-grade hardware. Targeting embedded systems, chip design, or hardware engineering roles.

Education

B.E. Electrical Engineering · National University of Sciences & Technology (NUST)

Sep 2022 – Jun 2026 (Expected)

CGPA: 3.79 / 4.00

Electronics, Automation & Energy Systems | EQF Level 6

- Multiple **Academic Excellence Awards** at NUST; consistently recognised as top student across cohort.
- Relevant coursework: VLSI Design, DSP, Analog Circuit Design, Control Systems, Power Electronics, Embedded Systems, RF & Microwave Engineering, Computer Architecture.

Work Experience

Electrical Engineer (Concurrent with B.E.) · RiseTech

Feb 2024 – Present

📍 Islamabad, Pakistan

- Signal Integrity & High-Speed PCB:** Engineered USB 2.0 differential pairs with **90 Ω controlled-impedance** routing on multi-layer boards for **EN 60601-1** medical devices; eliminated EMI failures in production.
- Production Hardware Lifecycle:** Delivered **4 production-ready systems** from schematic capture (EasyEDA / Cadence Allegro) through DFM review to JLCPCB manufacturing handoff.
- Firmware – STM32 & Renesas:** Designed bare-metal and RTOS-based firmware in STM32CubeIDE and e2 studio; implemented PID control loops and optimised ISR latency below **5 μs**.
- Power Domain Engineering:** Designed ESP32 architectures spanning **100 mA** sleep to **12 A DC** peak load; validated stability with LTSpice transient simulations.
- Protocol Stack:** Implemented BLE 5.0, Wi-Fi 802.11 b/g/n, SPI, I²C, USB 2.0, Wiegand, and RFID drivers from datasheet specifications.
- Maintained **100 % on-time project delivery** on all deadlines while sustaining a 3.79 CGPA concurrently.

Embedded System Designer (Internship) · Research Instruments & Measurement Systems (RIMS)

Jun 2025 – Aug 2025

📍 Islamabad, Pakistan

- Avionics Test Engineering:** Built Avionics LRU and Portable Tester ATP applications in Python for **JF-17 Thunder** maintenance; deployed signal-monitoring and visualisation dashboards in live maintenance operations.
- RF & SDR Pipelines:** Architected real-time SDR interfaces on **USRP X310** with GNU Radio for military-protocol visualisation; handled signals up to **5 GHz** at **580 Mbps**.
- Industrial DAQ:** Programmed NI myRIO & NI myDAQ in LabVIEW; achieved sub-**1 %** sensor-calibration error in precision acquisition architectures.
- PLC & Network Integration:** Developed TCP/UDP APIs for Arduino Opta Embedded Controller; interfaced with LabVIEW panels for real-time monitoring.

Key Projects

Underground Vehicle Profiling System – Edge AI (Final Year Project)

Mar 2025 – Present

- 95 % real-time classification accuracy** via sensor fusion and TensorFlow Lite Edge-AI inference on localised embedded hardware.
- Dual-firmware (V1/V2) pipeline for high-speed SD logging and edge processing improved data throughput by **40 %**.

EPIC – Real-Time Epilepsy Detection Device

Feb 2025 – Present

- On-device ML inference improved seizure-prediction accuracy by **30 %**; hardware alert path reduced notification latency by **25 %**.

EKKO Medical Device (Portable & Clinical) – RiseTech

Feb 2024 – Present

- Maintained **100 % EN 60601-1 compliance**; redesigned PCB stack-up reducing manufacturing defects by **15 %**.

Multi-Gas AQI Monitor – 9-Variable Atmospheric Sensor Node

Mar 2024 – Sep 2024

- STM32-based hub tracking CO₂, SO₂, PM2.5 and 6 further variables; calibration error margin **< 5 %**.

Ergonomic Mouse – Custom BLE PCB

Sep 2024 – Nov 2024

- Compact internal PCB with BLE 5.0 and dual-joystick control within tight ergonomic form-factor constraints.

Technical Skills

Microcontrollers & SoC: STM32 (F/G/H), ESP32/ESP8266, Renesas RA, PIC, NI myRIO/myDAQ, Arduino Opta PLC

Firmware & RTOS: Bare-metal C/C++, FreeRTOS, STM32CubeIDE, e2 studio, HAL/LL drivers, ISR optimisation, DMA

PCB & Hardware Design: Cadence Allegro X, KiCad, Altium Designer, EasyEDA; multi-layer PCB, controlled-impedance routing, DFM/DFT

Signal Integrity & High-Speed: USB 2.0 (90 Ω diff pairs), DDR trace matching, EMI mitigation, LTSpice SI analysis

Analog / Mixed-Signal IC Design: CMOS analog design, Cadence Virtuoso schematic & layout, post-layout simulation, parasitic extraction

RF & SDR: USRP X310, GNU Radio, spectral analysis, filter design, 5 GHz BW, military-protocol visualisation

Communication Protocols: USB 2.0, BLE 5.0, Wi-Fi 802.11, SPI, I²C, UART, TCP/UDP, Wiegand, RFID, gRPC

Programming & Tools: C/C++ (expert), Python (advanced), MATLAB/Simulink, LabVIEW, LaTeX, Git/GitHub

AI / Edge Computing: TensorFlow Lite, Edge AI inference, ML pipeline design, sensor fusion

Honours, Awards & Leadership

AirTech 24 – 1st Place			Robotics Roboline, AIR University; PID & DSP for path optimisation	Mar 2024
Academic Awards	Excellence		Multiple NUST awards; top student across cohort, 2022–Present	Ongoing
	President, EE Student Body		Led 400+ student council 2025-26; organised VoltFest and all events	Sep 2025–Present
		EasyEDA Campus Ambassador		Official NUST rep promoting PCB design education
CMC Mentor			NUST CMC mentorship programme – industry insights & development	Ongoing

Certifications

Advanced Gigabit Channel Design – Teledyne LeCroy	Sep 2025
High-Speed Digital Design – Teledyne LeCroy	Jul 2025
Cadence Design Systems: Essential Guide – Coursera	Jun 2025
AI For Everyone – Coursera / DeepLearning.AI	Jun 2025
MATLAB Onramp – MathWorks	Apr 2024

Additional

Languages: Urdu (Native), English (C2 Reading / B2 Speaking & Writing)

Sports: Cricket captain – college & departmental teams; represent department in 4 annual inter-departmental events.

Interests: Robotics, VLSI & Chip Design, Edge AI, RF Systems, Open-Source Hardware.